

Max-Splitting in the IID k -Choice Prophet Problem: A Corrected Analysis and Rigorous Partial Progress

June 23, 2026

Status. This note does *not* solve the finite-horizon question for $k > 1$. The proposed divisibility proof cannot be repaired from its stated ingredients: its “merging certificate” is equivalent to the desired theorem and is not proved by the cited paper. The rigorous progress obtained here is a block-compression theorem with an explicit duplicate-selection defect, an exact proof of the splitting argument for $k = 1$, and exact obstructions showing why the same argument does not close for $k > 1$.

Abstract

For a nonnegative distribution F , let $\rho_{n,k}(F)$ be the ratio between the optimal online value with n IID observations and k selections and the expected sum of the largest k observations. Let $\gamma_{k,n} = \inf_F \rho_{n,k}(F)$. Jiang, Ma, and Zhang ask whether, for each $k > 1$, the asymptotic value equals $\inf_n \gamma_{k,n}$. A proposed proof asserts the divisibility inequality $\gamma_{k,mn} \leq \gamma_{k,n}$ by an “equal-split merging theorem.” We show that the displayed merging certificate is merely an equivalent reformulation of divisibility and is not a theorem in the cited work of Brüstle, Pérez-Salazar, and Verdugo.

We then develop a valid replacement. If $G = F^{1/m}$ and mn IID G -observations are divided into n blocks of size m , their block maxima are IID with law F . We prove that the fine online value is at most the coarse online value plus the expected sum of the largest $k - 1$ residual observations left after deleting one maximum from every block. The same residual controls the increase in the prophet value. This yields an explicit “defective divisibility” inequality and gives exact divisibility when $k = 1$. For $k > 1$ the defect need not vanish; a two-point example has a strictly positive limiting defect. We also give an exact finite example in which max-splitting increases the pointwise competitive ratio. Consequently, a full solution requires a genuinely ratio-level charging argument for duplicate selections. The present results are partial progress only.

1 The problem and the status of the proposed proof

Let $X_1, \dots, X_n$ be IID with a nonnegative Borel distribution F . For a finite multiset $z = (z_1, \dots, z_s)$, let $\text{Top}_r(z)$ denote the sum of its r largest entries, with the conventions that $\text{Top}_0(z) = 0$ and that all entries are summed if $r \geq s$. Define

$$\begin{aligned} \text{OPT}_{n,k}(F) &:= \mathbb{E}[\text{Top}_k(X_1, \dots, X_n)], \\ \text{ALG}_{n,k}(F) &:= \sup_{\pi} \mathbb{E} \left[\sum_{i \in S_{\pi}} X_i \right], \\ \rho_{n,k}(F) &:= \frac{\text{ALG}_{n,k}(F)}{\text{OPT}_{n,k}(F)}, \quad \gamma_{k,n} := \inf_F \rho_{n,k}(F), \end{aligned}$$

where π ranges over adapted policies selecting at most k observations, and distributions with zero or infinite prophet value are omitted.

Jiang, Ma, and Zhang [1] give an exact semi-infinite linear program for $\gamma_{k,n}$ and explicitly record, for $k > 1$, the question

$$\lim_{N \rightarrow \infty} \gamma_{k,N} \stackrel{?}{=} \inf_{N \geq 1} \gamma_{k,N}. \quad (1)$$

Recent work continues to treat the exact multiple-selection ratio as unresolved; see, for example, [2, 3]. The arguments below are stated with $\liminf$ and therefore do not require a separate assumption about existence of the full limit.

1.1 The Type-Coverage formulation

For $N, k \geq 1$, set

$$B_{N,k}(q) := \mathbb{E}[\min\{\text{Bin}(N, q), k\}], \quad q \in (0, 1].$$

The Type-Coverage flow polytope $\mathcal{P}_N^k$ consists of nonnegative variables (x_i^ℓ, y_i^ℓ) satisfying

$$\begin{aligned} x_1^k &= 1, & x_1^\ell &= 0 \quad (\ell < k), \\ 0 &\leq y_i^\ell \leq x_i^\ell, \\ x_{i+1}^\ell &= x_i^\ell - y_i^\ell + y_i^{\ell+1}, \end{aligned}$$

with the standard zero boundary variables. Its coverage curve is

$$C_{x,y}(q) := \sum_{i=1}^N \sum_{\ell=1}^k \min\{y_i^\ell, qx_i^\ell\}.$$

The exact characterization of Jiang, Ma, and Zhang is

$$\gamma_{k,N} = \max_{(x,y) \in \mathcal{P}_N^k} \inf_{q \in (0,1]} \frac{C_{x,y}(q)}{B_{N,k}(q)}. \quad (2)$$

For later clarity, define the set of feasible guarantee levels

$$\mathcal{L}_{N,k} := \left\{ \beta \geq 0 : \begin{array}{l} \text{there exists } (x, y) \in \mathcal{P}_N^k \text{ such that} \\ C_{x,y}(q) \geq \beta B_{N,k}(q) \text{ for every } q \in (0, 1] \end{array} \right\}.$$

By (2), compactness of the finite flow polytope, and downward closure in β ,

$$\mathcal{L}_{N,k} = [0, \gamma_{k,N}]. \quad (3)$$

Proposition 1 (The asserted certificate is the desired theorem). *For fixed m, n, k , the implication*

$$\beta \in \mathcal{L}_{mn,k} \implies \beta \in \mathcal{L}_{n,k} \quad \text{for every } \beta \geq 0 \quad (4)$$

is equivalent to

$$\gamma_{k,mn} \leq \gamma_{k,n}.$$

Proof. By (3), implication (4) says exactly that $[0, \gamma_{k,mn}] \subseteq [0, \gamma_{k,n}]$, which is equivalent to the stated numerical inequality. $\square$

Thus the proposed manuscript’s key displayed “merging certificate” does not follow from its preceding calculations: it contains the entire divisibility claim. Moreover, the currently posted version of [2] develops an exact infinite-dimensional program and asymptotic lower bounds, but does not state or prove (4). A valid proof must explicitly construct a single coarse flow, independent of q , and verify all flow and coverage inequalities simultaneously.

2 Max-splitting and two elementary obstructions

For an integer $m \geq 1$, define the max-root of F by

$$G_m(x) := F(x)^{1/m}. \quad (5)$$

If $Y_1, \dots, Y_m$ are IID with law G_m , then

$$\max_{1 \leq j \leq m} Y_j \sim F.$$

This identity is correct, but it does not preserve the k -unit prophet problem when $k > 1$.

Proposition 2 (The rank benchmark is not preserved). *Let $q \in (0, 1)$ and put*

$$r = 1 - (1 - q)^{1/m}.$$

If $T \sim \text{Bin}(mn, r)$ counts successful components and $Z \sim \text{Bin}(n, q)$ counts blocks containing at least one success, then one can couple them so that $Z \leq T$ almost surely. Consequently,

$$B_{n,k}(q) \leq B_{mn,k}(r), \quad (6)$$

and the inequality can be strict.

Proof. Partition the mn Bernoulli trials into n blocks of size m . A block is nonempty with probability $1 - (1 - r)^m = q$, independently across blocks, so its number of nonempty blocks has law Z . The number of nonempty blocks never exceeds the total number of successes, hence $\min\{Z, k\} \leq \min\{T, k\}$ pathwise. Taking expectations proves (6). For $m = 2, n = 1, k = 2, q = 1/2$, one has $r = 1 - 1/\sqrt{2}$ and

$$B_{1,2}(1/2) = \frac{1}{2} \quad \text{whereas} \quad B_{2,2}(r) = 2 - \sqrt{2} > \frac{1}{2}.$$

□

The online numerator also need not contract separately. With two selection slots, two IID Bernoulli- p unit-valued components, and zero continuation value, the fine policy accepts both successes and obtains expectation $2p$. A single observation equal to their maximum has expectation $1 - (1 - p)^2 = 2p - p^2 < 2p$. Thus a Bellman comparison that simply replaces a block by its maximum is false for $k > 1$.

The stronger pointwise ratio comparison is false as well.

Proposition 3 (Exact counterexample to pointwise ratio contraction). *Let $n = 3, k = 2, m = 2$, and let F be supported on $\{0, 1, 11/10\}$ with probabilities*

$$\left(\frac{1}{30}, \frac{2}{15}, \frac{5}{6} \right).$$

Let $G = F^{1/2}$. Then

$$\rho_{6,2}(G) > \rho_{3,2}(F). \quad (7)$$

Numerically, the two ratios are approximately 0.99493608 and 0.99431458, respectively.

Proof. The exact backward-induction calculation in Appendix A gives

$$\begin{aligned} \text{ALG}_{3,2}(F) &= \frac{1567}{720}, & \text{OPT}_{3,2}(F) &= \frac{13133}{6000}, \\ \text{ALG}_{6,2}(G) &= \frac{4741}{2160} - \frac{53\sqrt{6}}{10800} - \frac{7\sqrt{30}}{18000} + \frac{17\sqrt{5}}{18000}, & \text{OPT}_{6,2}(G) &= \frac{1101}{500} - \frac{\sqrt{6}}{360} - \frac{\sqrt{30}}{4500}. \end{aligned}$$

Since both prophet values are positive, (7) is equivalent to positivity of the cross-product difference. Direct simplification gives

$$\begin{aligned} &\text{ALG}_{6,2}(G) \text{OPT}_{3,2}(F) - \text{ALG}_{3,2}(F) \text{OPT}_{6,2}(G) \\ &= \frac{3848525 + 669783\sqrt{5} - 119093\sqrt{30} - 1521495\sqrt{6}}{324000000} > 0. \end{aligned}$$

For a fully rational sign check, use $\sqrt{5} > 2.23$, $\sqrt{6} < 2.45$, and $\sqrt{30} < 5.48$; the numerator is then larger than 961000. $\square$

Proposition 3 does not disprove $\gamma_{k,mn} \leq \gamma_{k,n}$, because the latter compares infima over all distributions. It does rule out a natural pointwise repair of the proposed argument.

3 A rigorous block-compression theorem

We now give a valid replacement for the missing merging step. It isolates exactly the contribution that is lost when several selections occur in one split block.

Fix F, m, n, k , let $G = F^{1/m}$, and couple mn IID G -observations as

$$(Y_{b,j} : b \in [n], j \in [m]).$$

For each block b , let

$$M_b := \max_{j \in [m]} Y_{b,j}.$$

Then $M_1, \dots, M_n$ are IID with law F . In each block choose a distinguished maximizing index, for example the smallest index attaining the maximum, and delete that one observation. Let $\mathcal{R}$ be the multiset of all remaining $n(m-1)$ observations. Define the residual defect

$$\Delta_{n,m,k}(F) := \mathbb{E}[\text{Top}_{k-1}(\mathcal{R})]. \quad (8)$$

Theorem 4 (Block compression with a residual defect). *Assume the relevant prophet values are finite. Then*

$$\text{ALG}_{mn,k}(F^{1/m}) \leq \text{ALG}_{n,k}(F) + \Delta_{n,m,k}(F), \quad (9)$$

$$\text{OPT}_{n,k}(F) \leq \text{OPT}_{mn,k}(F^{1/m}) \leq \text{OPT}_{n,k}(F) + \Delta_{n,m,k}(F). \quad (10)$$

Proof. We first prove (9). Fix any adapted fine policy π that selects at most k of the mn observations. A policy for the coarse sequence $M_1, \dots, M_n$ can simulate π as follows. On observing M_b , it samples an m -tuple from the regular conditional law of $(Y_{b,1}, \dots, Y_{b,m})$ given its maximum M_b , independently of the conditional samples used in other blocks, and advances the virtual state of π through that block. It accepts M_b if and only if the virtual policy selects at least one component in block b . This is an adapted randomized coarse policy. It accepts at most k block maxima, because the number of occupied blocks is at most the total number of virtual selections. (For finite-support distributions

the conditional simulation is elementary; for general Borel laws it follows from existence of regular conditional probabilities.)

Let S_b be the indices selected by π in block b , and let $I_b = \mathbf{1}\{S_b \neq \emptyset\}$. In every occupied block choose an anchor as follows: if the distinguished maximizing index is selected, use it; otherwise use any selected index. The anchor value is at most M_b , and all other selected indices belong to the residual multiset $\mathcal{R}$. Since there is one anchor per occupied block, the number of nonanchor selections is at most

$$k - \sum_{b=1}^n I_b \leq k - 1$$

whenever any selection is made. Hence, pathwise,

$$\sum_{b=1}^n \sum_{j \in S_b} Y_{b,j} \leq \sum_{b=1}^n I_b M_b + \text{Top}_{k-1}(\mathcal{R}). \quad (11)$$

The first term on the right is the reward of the simulated coarse policy, so its expectation is at most $\text{ALG}_{n,k}(F)$. Taking expectations in (11), and then taking the supremum over fine policies, proves (9).

For the lower bound in (10), the distinguished block maxima form a submultiset of the fine observations, so

$$\text{Top}_k(M_1, \dots, M_n) \leq \text{Top}_k((Y_{b,j})_{b,j})$$

pathwise. For the upper bound, apply the same anchor construction to a set of indices attaining the k largest fine observations. The anchors contribute at most $\text{Top}_k(M_1, \dots, M_n)$, and the nonanchors contribute at most $\text{Top}_{k-1}(\mathcal{R})$. Taking expectations proves the second inequality. $\square$

Corollary 5 (Defective divisibility). *For every F with positive finite prophet value,*

$$\rho_{mn,k}(F^{1/m}) \leq \rho_{n,k}(F) + \frac{\Delta_{n,m,k}(F)}{\text{OPT}_{mn,k}(F^{1/m})}, \quad (12)$$

$$\gamma_{k,mn} \leq \inf_F \left\{ \rho_{n,k}(F) + \frac{\Delta_{n,m,k}(F)}{\text{OPT}_{mn,k}(F^{1/m})} \right\}. \quad (13)$$

In particular, for $k = 1$,

$$\gamma_{1,mn} \leq \gamma_{1,n}. \quad (14)$$

Proof. By theorem 4,

$$\rho_{mn,k}(F^{1/m}) \leq \frac{\text{ALG}_{n,k}(F) + \Delta_{n,m,k}(F)}{\text{OPT}_{mn,k}(F^{1/m})}.$$

The lower bound in (10) implies

$$\frac{\text{ALG}_{n,k}(F)}{\text{OPT}_{mn,k}(F^{1/m})} \leq \frac{\text{ALG}_{n,k}(F)}{\text{OPT}_{n,k}(F)} = \rho_{n,k}(F),$$

which proves (12). Taking the infimum over F proves (13). If $k = 1$, then $\text{Top}_{k-1} = \text{Top}_0 = 0$, so the defect vanishes. Taking the infimum in the resulting pointwise inequality gives (14). $\square$

Thus the elementary max-splitting proof is complete for one selection. For multiple selections, the only term not controlled at the required ratio is the value of duplicate selections from the same block.

4 The residual defect need not vanish

One might hope that, for fixed F, n, k , the defect in (12) tends to zero as $m \rightarrow \infty$. This is false even for a bounded two-point distribution.

Proposition 6 (Persistent duplicate mass). *Let F put probability $1/2$ on each of 0 and 1 , and take $k = 2$. Then*

$$\lim_{m \rightarrow \infty} \Delta_{n,m,2}(F) = 1 - \left(\frac{1 + \log 2}{2} \right)^n > 0. \quad (15)$$

Proof. Under $G_m = F^{1/m}$, a component equals 1 with probability

$$p_m = 1 - 2^{-1/m}.$$

Let $Z_{b,m} \sim \text{Bin}(m, p_m)$ be the number of ones in block b . After one block maximum is deleted, the residual multiset contains a one from block b exactly when $Z_{b,m} \geq 2$. Since $k - 1 = 1$,

$$\Delta_{n,m,2}(F) = \mathbb{P} \left(\max_{1 \leq b \leq n} Z_{b,m} \geq 2 \right) = 1 - \mathbb{P}(Z_{1,m} \leq 1)^n.$$

Now $mp_m \rightarrow \log 2$, so $Z_{1,m}$ converges in distribution to $\text{Pois}(\log 2)$. Therefore

$$\mathbb{P}(Z_{1,m} \leq 1) \rightarrow e^{-\log 2}(1 + \log 2) = \frac{1 + \log 2}{2},$$

which proves (15). □

For this particular two-point law the actual competitive ratio is one, so theorem 6 is not a counterexample to (1). It is a counterexample to the claim that duplicate mass disappears automatically under finer max-splitting. Any successful proof for $k > 1$ must charge that mass against the corresponding increase in the prophet benchmark, rather than bounding it by an additive $o(1)$ term.

5 What remains to obtain a full solution

The preceding theorem identifies a precise missing inequality. For a fine policy π , let H_π be the expected reward of the compressed coarse policy constructed in the proof of theorem 4, and let

$$G_{\text{prop}} := \text{OPT}_{mn,k}(F^{1/m}) - \text{OPT}_{n,k}(F)$$

be the prophet's gain from retaining all split components. Since $H_\pi \leq \text{ALG}_{n,k}(F)$, a sufficient ratio-level charging inequality would be

$$\text{ALG}_{mn,k}(F^{1/m}) - H_\pi \leq \rho_{n,k}(F) G_{\text{prop}} \quad (16)$$

for an optimal or asymptotically optimal fine policy π . Indeed, (16) would imply

$$\begin{aligned} \text{ALG}_{mn,k}(F^{1/m}) &\leq \text{ALG}_{n,k}(F) + \rho_{n,k}(F) (\text{OPT}_{mn,k}(F^{1/m}) - \text{OPT}_{n,k}(F)) \\ &= \rho_{n,k}(F) \text{OPT}_{mn,k}(F^{1/m}), \end{aligned}$$

and hence pointwise ratio contraction. Proposition 3 shows that this particular pointwise inequality cannot hold for all F . At the worst-case level, the required statement is instead the Type-Coverage

implication (4). Establishing that implication would solve the problem; merely naming it a merging theorem does not.

A weaker but still sufficient route would be to show, for every fixed n , that distributions arbitrarily close to minimizing $\rho_{n,k}$ can be chosen so that the normalized defect in (12) tends to zero along arbitrarily large m . Corollary 5 would then yield

$$\liminf_{m \rightarrow \infty} \gamma_{k,mn} \leq \gamma_{k,n},$$

and, if the full asymptotic limit exists, equality in (1) would follow by taking the infimum over n . Proposition 6 shows that such a residual-negligibility statement is not true for arbitrary fixed distributions, so it would have to exploit structure of near-worst-case instances.

6 Conclusion

The original manuscript is directly aimed at the Jiang–Ma–Zhang open problem, but its central step is unproved and is equivalent to the desired result. The correct max-splitting analysis yields the exact inequalities (9)–(13). These recover the familiar one-selection divisibility argument and rigorously isolate the obstruction for $k > 1$: multiple useful components can occur in one split block, and their contribution persists in the fine-splitting limit. The exact pointwise counterexample further rules out a distribution-by-distribution contraction. Accordingly, the outcome of this note is **partial progress, not a solution**. A full proof still requires a new ratio-level duplicate-charging or Type-Coverage merging argument.

A Exact calculation for Proposition 3

For a law H and integers $s, \ell \geq 0$, let $V_{s,\ell}(H)$ denote the optimal online value with s IID observations remaining and ℓ selections remaining. The standard recursion is

$$V_{0,\ell}(H) = V_{s,0}(H) = 0, \quad V_{s,\ell}(H) = \mathbb{E}[\max\{V_{s-1,\ell}(H), X + V_{s-1,\ell-1}(H)\}]. \quad (17)$$

Thus a value x is accepted exactly when $x \geq V_{s-1,\ell}(H) - V_{s-1,\ell-1}(H)$.

For the law F in Proposition 3, direct substitution in (17) gives the following acceptance sets; b denotes $11/10$.

s	$\ell = 1$	$\ell = 2$
1	$\{0, 1, b\}$	$\{0, 1, b\}$
2	$\{b\}$	$\{0, 1, b\}$
3	$\{b\}$	$\{b\}$

The resulting final state is

$$V_{3,2}(F) = \frac{1567}{720}.$$

The square-root law $G = F^{1/2}$ has probabilities

$$\mathbb{P}_G(0) = \frac{\sqrt{30}}{30}, \quad \mathbb{P}_G(1) = \frac{\sqrt{6}}{6} - \frac{\sqrt{30}}{30}, \quad \mathbb{P}_G(b) = 1 - \frac{\sqrt{6}}{6}. \quad (18)$$

The exact threshold comparisons in (17) give

s	$\ell = 1$	$\ell = 2$
1	$\{0, 1, b\}$	$\{0, 1, b\}$
2	$\{1, b\}$	$\{0, 1, b\}$
3	$\{b\}$	$\{1, b\}$
4	$\{b\}$	$\{1, b\}$
5	$\{b\}$	$\{b\}$
6	$\{b\}$	$\{b\}$

Substitution of (18) then yields

$$V_{6,2}(G) = \frac{4741}{2160} - \frac{53\sqrt{6}}{10800} - \frac{7\sqrt{30}}{18000} + \frac{17\sqrt{5}}{18000}.$$

All comparisons are strict except for the immaterial choice of accepting a zero in the last period. They can be checked using the rational enclosures $2.23 < \sqrt{5} < 2.24$, $2.44 < \sqrt{6} < 2.45$, and $5.47 < \sqrt{30} < 5.48$.

Finally, for any law supported on $\{0, 1, b\}$,

$$\text{OPT}_{s,2}(H) = B_{s,2}(\mathbb{P}_H(X \geq 1)) + (b-1)B_{s,2}(\mathbb{P}_H(X \geq b)).$$

Applying this identity to F and G gives the prophet values displayed in the proof of Proposition 3.

References

- [1] J. Jiang, W. Ma, and J. Zhang, Tightness without counterexamples: a new approach and new results for prophet inequalities, arXiv:2205.00588, current version 2025, <https://arxiv.org/abs/2205.00588>.
- [2] J. Brüstle, S. Pérez-Salazar, and V. Verdugo, Splitting guarantees for prophet inequalities via nonlinear systems, arXiv:2406.17767, current version 2025, <https://arxiv.org/abs/2406.17767>.
- [3] E. Cruz-Ossa, S. Pérez-Salazar, and V. Verdugo, Competition versus complexity in multiple-selection prophet inequalities, arXiv:2602.20398, 2026, <https://arxiv.org/abs/2602.20398>.
- [4] P. Nuti, Open Problems in Operations Research, problem W4409922425_p0, https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4409922425_p0.